Module 5: A+ Understanding and Maintenance of Networks

DAY 5:-

Task 1:

```
aashanapatel@192 ~ % ping 192.168.1.4
PING 192.168.1.4 (192.168.1.4): 56 data bytes
64 bytes from 192.168.1.4: icmp_seq=0 ttl=64 time=0.107 ms
64 bytes from 192.168.1.4: icmp_seq=1 ttl=64 time=0.295 ms
64 bytes from 192.168.1.4: icmp_seq=2 ttl=64 time=0.187 ms
64 bytes from 192.168.1.4: icmp_seq=3 ttl=64 time=0.249 ms
64 bytes from 192.168.1.4: icmp_seq=4 ttl=64 time=0.256 ms
64 bytes from 192.168.1.4: icmp_seq=5 ttl=64 time=0.241 ms
64 bytes from 192.168.1.4: icmp_seq=6 ttl=64 time=0.196 ms
```

Task 2:

```
aashanapatel@192 ~ % traceroute www.youtube.com
traceroute: Warning: www.youtube.com has multiple addresses; using 142.251.151.4
traceroute to www.youtube.com (142.251.151.4), 64 hops max, 52 byte packets
 1  gpon.net (192.168.1.1)  3.904 ms  3.512 ms  4.898 ms
 2  10.130.192.1 (10.130.192.1)  4.221 ms  4.791 ms  4.986 ms
```

Task 3:

```
aashanapatel@192 ~ % ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
Request timeout for icmp_seq 4
Request timeout for icmp_seq 5
Request timeout for icmp_seq 6
ping: sendto: No route to host
Request timeout for icmp_seq 7
ping: sendto: Host is down
Request timeout for icmp_seq 8
ping: sendto: Host is down
Request timeout for icmp_seq 9
ping: sendto: Host is down
```

Task 4:

IP Address of the devices connected to my Home Network
192.168.1.2. - iPhone
192.168.1.3 - Macbook Pro
192.168.1.4. - iPhone

Task 5:

I. Network Setup Steps

1. **Connected the modem to the ISP line -** The modem was plugged into the incoming coaxial/fiber line from the internet service provider and powered on, establishing a connection to the ISP.

2. **Connected the router to the modem -** An Ethernet cable was run from the modem's output port to the router's **WAN port**. The router was powered on and allowed to obtain a public IP address via DHCP from the ISP.

3. **Connected a switch to the router -** To support more wired devices than the router's built-in LAN ports allow, an 8-port switch was connected to one of the router's **LAN ports** using a straight-through Ethernet cable.

4. **Connected wired devices to the switch -** Desktop PCs were connected to the switch's LAN ports using Ethernet cables.

5. **Connected wireless devices -** A laptop and a smartphone joined the network over Wi-Fi, broadcast by the router's built-in access point.

6. **Verified IP assignment -** Each device was confirmed to have received a valid local IP address (e.g., in the 192.168.1.x range) automatically via the router's DHCP server, with no duplicate-IP conflicts observed.

II. Connectivity Testing

Testing with ping

```
aashanapatel@192 ~ % ping 192.168.1.4
PING 192.168.1.4 (192.168.1.4): 56 data bytes
64 bytes from 192.168.1.4: icmp_seq=0 ttl=64 time=0.107 ms
64 bytes from 192.168.1.4: icmp_seq=1 ttl=64 time=0.295 ms
64 bytes from 192.168.1.4: icmp_seq=2 ttl=64 time=0.187 ms
64 bytes from 192.168.1.4: icmp_seq=3 ttl=64 time=0.249 ms
64 bytes from 192.168.1.4: icmp_seq=4 ttl=64 time=0.256 ms
64 bytes from 192.168.1.4: icmp_seq=5 ttl=64 time=0.241 ms
64 bytes from 192.168.1.4: icmp_seq=6 ttl=64 time=0.196 ms
```

Testing with traceroute / tracert

```
aashanapatel@192 ~ % traceroute www.youtube.com
traceroute: Warning: www.youtube.com has multiple addresses; using 142.251.151.4
traceroute to www.youtube.com (142.251.151.4), 64 hops max, 52 byte packets
 1  gpon.net (192.168.1.1)  3.904 ms  3.512 ms  4.898 ms
 2  10.130.192.1 (10.130.192.1)  4.221 ms  4.791 ms  4.986 ms
```